CACFTNet: A Hybrid Cov-Attention and Cross-Layer Fusion Transformer Network for Hyperspectral Image Classification

Shuli Cheng, Runze Chan, and Anyu Du

***Abstract*— Hyperspectral (HS) image classification has become an important research area. Although previous work on HS image classification has achieved impressive results, finding a proper balance between extracting spatial–spectral information and capturing band similarities remains a challenge. To address this problem, we designed some efficient modules and constructed hybrid covariance attention and cross-layer fusion transformer network (CACFTNet), which efficiently models the extraction of spatial–spectral information and band similarity. First, our approach combines local and global perspectives to process the desired feature information. We introduce the dual-branch feature processing (DBFP) module, which can model the desired spatial–spectral information and band similarity. Second, we design the dual-branch feature fusion (DBFF) module, which combines a convolutional neural network (CNN) and a transformer to catch spatial–spectral information at different scales and fuse them effectively. To further process the features, we construct the unparameterized covariance attention (UPCA) module, which utilizes the covariance matrix to capture the correlation between different spectral bands. This allows the network to concentrate on bands that are more useful for classification tasks. In addition, we designed a hybrid activation function (HAF) that maps the channel values to specific ranges and emphasizes the extent to which the correlation varies between different bands. Finally, in order to incorporate important information from different layers, we propose the cross-layer adaptive attention fusion (CAAF) module, which fully fuses information between layers and enriches the overall information representation. We evaluate our proposed model on three well-known public datasets and demonstrate its superiority over existing approaches.**

***Index Terms*— Covariance, cross-layer attention, feature fusion, hyperspectral (HS) image classification, transformer.**

I. Introduction

HYPERSPECTRAL (HS) images are obtained through imaging with a spectrometer, possessing rich spectral and spatial characteristics. Compared to traditional red, green, blue (RGB) images, HS images contain a wider range of information in continuous spectral bands. This makes HS images play an irreplaceable and important role in various fields. So far, HS images have been successfully applied in multiple domains, such as environmental monitoring [1], precision agriculture detection [2], earthquake prediction [3], and have achieved significant results.

In the early stages of the research, the main focus was on spectral features in order to obtain the desired classification results. This includes the use of traditional methods such as K-nearest neighbor (KNN) [4], support vector machine (SVM) [5], and random forest (RF) [6], and so on. Although these methods perform well in terms of performance, their classification performance is still unsatisfactory. The features extracted by traditional methods have the disadvantages of poor generalization and robustness and low expressive power.

In recent years, with the continuous development of deep learning (DL), numerous high-performing networks have been proposed, including stacked autoencoders [7], deep belief networks [8], graph convolutional networks (GCNs) [9], convolutional neural networks (CNNs) [10], and visual transformers (ViTs) [11]. Compared to the traditional methods mentioned above, DL-based approaches enable efficient end-to-end feature extraction and the learning of high-level semantic information. For example, Wang et al. [12] proposed a novel uncertainty-guided trustworthy fusion network (UTFNet) for RGB-T semantic segmentation. This fusion network takes into account the fact that dynamically estimating the uncertainty of each modality for different samples can help the model perceive the changes in the information quality, thus providing guidance for reliable fusion.

CNNs are one of the most popular methods in DL. They model the local spectral and spatial features of HS images and utilize weight-sharing mechanisms to enhance the discriminative power of features. In order to extract spectral feature information from HS images, 1D-CNN [13] has been proposed to capture more abstract feature representations by performing efficient local feature extraction through multilayer convolution. However, 1-D convolution ignores the rich spatial information in HS images. Therefore, Ghaderizadeh et al. [14] proposed a method that combines 2-D and 3-D convolutions to extract both spectral and spatial features, using 2-D convolution for better identification of spatial characteristics.

Although the CNN-based networks proposed by the aforementioned researchers have shown excellent performance

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in spectral–spatial feature extraction, it is challenging to effectively utilize and integrate both spectral and spatial information simultaneously through single-scale analysis. Furthermore, the ability to extract information solely relying on convolutional layers is limited, as the receptive field is restricted. Therefore, the researchers proposed a series of innovative multiscale two-branch networks to solve this problem. In order to efficiently capture multiscale spectral–spatial information, [15] constructed multiarea target attention (MATA) by combining a shared feature extractor (FE) and a classifier with a class decision weighting strategy. The method is better able to take into account the fact that objects corresponding to different classes of pixels exhibit different size features. Duan et al. [16] proposed a model called structure-preserving hypergraph convolutional network (SPHGCN) that combines the characteristics of CNN and GCN. They constructed two high-order convolution branches starting from pixel-level samples and patch samples to span multiple domains and learn common information. In addition, they broke through the boundary limitations of spatial convolution and achieved semantic feature extraction across HS image blocks. Zhang et al. [17] proposed a fusion network called interleaved perception CNN (IP-CNN) for integrating HS data and light detection and ranging (LiDAR) data. The network takes the fused data as input and performs the final classification using a dual-branch CNN. In [18], a dual-branch dual-attention mechanism is developed by combining spatial attention and channel attention. This mechanism adaptively adjusts the importance of different locations and channels to better capture details and critical information in an image. Guo et al. [19] designed a dual-spectral and global spatial feature fusion network (DSGSF) from both the spectral and spatial perspectives. It includes a spatial subnetwork and a spectral subnetwork to extract the spatial and spectral features of the image, respectively. Then, the features extracted from the two subnetworks are fused through a fusion module to achieve a more comprehensive feature representation and more accurate classification results. In addition, in order to fully leverage the advantages of CNN and GCN, Zhou et al. [20] proposed a multifeature fusion network. This network consists of a CNN subnetwork and a GCN subnetwork, which extract features from two different scales. Finally, through an attention fusion module, the features from different scales are fused. However, most of the above dual-branch structures mainly focus on the extraction of spectral features and spatial features, without fully considering the similarity relationship between different bands. In addition, some networks only focus on extracting local features, neglecting the importance of global information.

Recently, transformer-based models have been widely used in HS image classification tasks. Compared to CNN, the transformer has a stronger learning ability and can model global information. ViT is the first model that introduces a pure transformer architecture into the field of computer vision. SpectralFormer [21] proposes a grouped spectral embedding based on the characteristics of the transformer, which learns local sequential information from adjacent bands of the HS image. It also designs cross-layer skip connections to fuse information from different levels. MorphFormer [22] designs two learnable spectral and spatial morphological networks that enable interaction between HS image tokens and classifiable tokens through morphological convolution. In order to fully consider different types of features, Zhao et al. [23] proposed a multiview attention hybrid network (MVAHN) that embeds CNN operations into ViT to capture both local and global features. This is done to overcome the limitations faced by convolutions in dealing with long-range feature interactions. In order to overcome the limitations of CNN's limited receptive field and ViT's inability to capture local information completely, Zhao et al. [24] proposed a network structure called fractional Fourier image transformer (FrIT). FrIT reduces the redundancy and loss of shallow and deep-layer information, effectively compensating for the deficiencies of both CNN and ViT. Zhao et al. [25] proposed a novel unsupervised domain adaptation (UDA) framework toward multilevel features and decision boundaries (ToMF-B) is proposed for the cross-scene hyperspectral image classification (HSIC), which can align task-related features and learn task-specific decision boundaries in parallel. Although the above transformer models perform well in capturing long-range dependencies, they often overlook the contribution of local features in HS images, leading to the loss of boundary information between categories. While some networks attempt to combine CNN with a transformer, they fail to effectively consider the interference between different scale features and do not separate the extraction of different scale features. In addition, they do not perform an effective fusion of these two different scale features to further enhance the expressive power of the features.

Furthermore, there is a significant correlation between spectral bands in HS images, which can provide valuable information ([26], [27], [28]). In order to fully consider the correlation information between different spectral bands, Zhang et al. [26] proposed a method that combines adaptive weighted filtering (AWF) and local covariance matrix representation (LCMR) to fully utilize the interplay between spectral and spatial information to improve classification performance. On the other hand, He et al. [28] used covariance maps to jointly leverage the spectral and spatial information of HS images, employing a multiscale strategy. This strategy ensures that different samples can benefit from it, enhancing the classification accuracy by enriching the information conveyed in the training samples. However, the covariance matrix, while capable of describing the correlation between different bands, cannot measure their differences and nonlinear relationships. By measuring differences, we can gain a clearer understanding of the similarities between different bands and highlight the information from different bands. If the difference is low, it may indicate high similarity, which could mean the presence of noise or a significant amount of redundant and repetitive information. On the other hand, if the difference is high, it may indicate low similarity, which could mean that the location has more prominent features. Therefore, in our approach, we not only consider the correlation but also focus on the measurement of differences to better reveal the information and features in the images.

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Therefore, in response to the above problem, we propose a novel hybrid covariance attention and cross-layer fusion transformer network (CACFTNet) for HS image classification. First, we recognize that both CNN and transformer methods have their respective advantages. To fully exploit their strengths, we employ a dual-branch feature fusion (DBFF) module to balance the capture of local and global spectral–spatial information. By summing the features, we fuse the features of the transformer branch with those of the CNN branch and use them as the feature information of the transformer branch to effectively merge features of different scales and enhance feature representation capability. Furthermore, to avoid excessive redundancy and repetitive information during the feature extraction phase, we introduce the unparameterized covariance attention (UPCA) module. This module can effectively measure the similarity information between different spectral bands without increasing the number of network parameters. In the CACFTNet algorithm, we also introduce a new hybrid activation function (HAF) that combines tanh and sigmoid nonlinear functions. Through the activation function of HAF, it can effectively compress and suppress redundant information in high-similarity bands, as well as noise in the image with high similarity, thereby enhancing the feature representation capability. In addition, inspired by references [21] and [29], we also designed the cross-layer adaptive attention fusion (CAAF) module, which is combined with the parameter-free attention module SimAm [30], to enable interaction of important information across different layers. Through this design, we avoid the introduction of unnecessary parameters to the network and reduce the loss of important information during the propagation process between layers. By fusing shallow and deep features, the performance of the transformer method is significantly improved, and more discriminative feature information is extracted from the data.

The specific contributions can be summarized as follows.

1) To facilitate effective interaction and fusion of features at different scales, we have designed a DBFF module. This module efficiently blends local and global spectral–spatial features to enhance the expressive power of the features.
2) To better explore the similarities between different bands and maximize the preservation of useful information while reducing noise and redundant information, we have introduced the UPCA module. This module is able to effectively describe the similarities between different bands and highlight the differences between them by comparing the covariance matrix differences of features captured by the CNN branch and the transformer branch. This emphasizes the information of each band. In addition, in order to further emphasize this differential feature as well as to enhance the nonlinear expression of the differential feature, we improve an HAF. This activation function contains both sigmoid and tanh nonlinear activation functions. The activation of the HAF function can make the positions in the dissimilarity matrix with higher dissimilarity have higher values after activation, while the positions with lower dissimilarity have lower values. This allows the network to focus more on the useful portion of the band information, which serves to reduce redundant information.
3) To achieve the interaction between low-level and high-level encoders and prevent the loss of important information during the propagation between layers, we propose a CAAF structure. This structure combines with the parameter-free attention mechanism called SimAm, which allows for the effective fusion of shallow and deep features without introducing additional network parameters. This fusion method fully utilizes the detailed information from shallow features while also leveraging the abstract representation capacity of deep features, further enhancing the model's expressive power and generalization ability.

The rest of the article is organized as follows. Section II briefly reviews the research on HS image classification in recent years. Section III describes the proposed network structure. Section IV shows experimental datasets, extensive ablation studies and discussions, and comparisons of classification accuracies. Section V gives specific conclusion.

II. Related Work

A. CNN-Based Classification Methods

When CNNs are employed for HS image classification tasks, the feature extraction mainly relies on the features within the CNN. The CNN automatically learns the features in the input data and progressively extracts higher-level features through multiple convolutional and pooling layers. Depending on the type of HS image features extracted, CNN models can be categorized into 1D-CNN, 2D-CNN, and 3D-CNN, as well as variants of CNN.

Early on in the task of HS image classification, Zhai et al. [31] proposed a 1-D CNN to extract feature information in the spectral domain of HS images. This CNN was designed to effectively extract local features using multilayer convolution and capture more abstract feature representations. However, 1-D convolution ignores the rich spatial information in HS images, so Ghaderizadeh et al. [14] proposed a network with a mixture of 2-D convolution and 3-D convolution (3D-2DCNN), which better jointly extracts informative representations of the spectral space and better recognizes the spatial features through 2-D convolution. However, as the depth of the network increases, the spatial–spectral information in HS images cannot be adequately extracted by considering only 1D-CNN or 3D-CNN. Therefore, in order to overcome these effects, Wang et al. [32] combined deep 3-D convolution and stochastic Fourier networks to improve the efficiency of classifying HS images. To reduce the parameters in the network and improve feature extraction for HS images, Zhang et al. [33] combined convolution and transformer to form a kind of two-branch network to capture HS features from local–global. In addition, to reduce the parameters in the network while extracting HS image features, Meng et al. [34] designed a lightweight spatial–spectral convolution to replace convolution. In addition, in order to better capture the nonlocal topological relationships of the underlying data structures of

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HS images while modeling the local spatial relationships of the samples, Zhang et al. [35] proposed a topological structure and semantic information transfer network (TSTnet). The method effectively integrates CNN and GCN, combining rich semantic information with topological structure information through the use of a graph constructed by CNN deep feature construction. Here, CNN extracts local spatial information while GCN extracts nonlocal spatial information.

These networks and their variations show excellent capabilities for capturing localized spatial information, but they often struggle to detect subtle differences between spectral bands. Their inability to model global contextual information and capture sequential data, especially when there are multiple spectrally similar feature classes, can lead to errors in identifying spectral classes. Therefore, some researchers have suggested resolving these issues by utilizing the global modeling capabilities of transformers.

B. Transformer-Based Classification Method

The emergence of transformers has made significant contributions to the field of computer vision. The transformer has a significant difference compared to CNN. CNN has unquestionable local modeling capabilities, while the transformer can effectively capture global information. This is attributed to its use of the self-attention mechanism module, which excels in handling sequential information.

In recent years, there has been a growing number of transformer-based variant networks, and many researchers have incorporated them into the HS image classification task, achieving very satisfactory classification accuracy. However, it also fails to effectively model local semantic information. Hong et al. [21] proposed a grouped spectral embedding based on the properties of the transformer to capture local sequence information from neighboring bands. They also designed a cross-layer hopping connection to integrate different levels of information. In order to better improve the classification effect in the case of small samples, Zhao et al. [36] proposed an HS classification framework based on multiattention transformer (MAT) and adaptive superpixel segmentation-based active learning (MAT-ASSAL). The framework is able to combine the transformer with the adaptive superpixel segmentation algorithm to achieve excellent classification results. In order to better delineate the boundaries of fine-grained categories, Zhang et al. [37] proposed a model known as the spatial logic aggregation network (SLA-NET), which utilizes morphological transformations for tree classification. The researchers developed two methods for feature extraction: spatial feature extraction and morphological feature extraction. Finally, these two features were fused adaptively. Liao et al. [38] proposed a spectral–spatial fusion transformer to obtain spatial–spectral features using a spectral transformer and spatial transformer separately. Finally, adaptive spatial–spectral feature fusion is performed on the extracted features to obtain advanced semantic features. Roy et al. [22] designed a spatial–spectral morphological attention transformer based on morphology. This transformer utilizes spectral and spatial morphological convolutions and integrates them with attention to enhance the interaction between the structural and shape information of HS image tokens and categorizable tokens. Zhao et al. [39] combined a CNN with a transformer to extract features from both local and global information using two fusion methods: summation and channel superposition. Due to the presence of multiple bands in HS images, the features extracted using the multihead self-attention mechanism (MHSA) in ViT may be too dispersed. To address this, Mei et al. [40] proposed the group-aware hierarchical transformer (GAHT), which confines the MHSA to local contexts by utilizing pixel grouping embedding, thus enabling the extraction of local–global features. Qiao et al. [41] designed a network that focuses on a specific pixel and all neighboring pixels within a chosen window of a particular size. The network extracts feature information at various scales by adjusting the sizes of the local windows. Liu et al. [42] extracted feature information from the center of HS image pixels and around the pixels using scaled dot product center of attention and central attention networks. Ren et al. [43] proposed an end-to-end inception transformer network (IFormer) that efficiently extracts spectral–spatial features from HS images and captures high and low-frequency information from the extracted features.

C. Classification Methods Based on Attention Mechanisms

The attention mechanism module serves as an effective enhancement module. On one hand, the introduction of the attention mechanism enables CNN feature extraction to prioritize more useful information and suppress irrelevant data, thereby enhancing the efficiency of feature extraction. On the other hand, the attention mechanism can enhance the model's capacity to assign weights to different features, allowing the model to more accurately capture the important features of the input data. Liang et al. [44] proposed a CNN architecture combined with transformer and SimAm attention [30]. This combination efficiently extracts features from both local and global perspectives. The incorporation of SimAm attention makes the network more attentive to important feature parts, thereby improving classification performance. Xu et al. [45] proposed a multiscale and cross-level attention learning (MCAL) network. This network first extracts multiscale information using a multiscale feature extraction module then utilizes a cross-level feature fusion module to merge information from different levels, and finally employs a spectral attention module to emphasize spatial context and spectral information. Paoletti et al. [46] proposed an automated design and optimization of CNNs based on the channel attention mechanism. More efficient CNNs are automatically generated from a small number of samples, and the most effective features in the spectral domain are extracted through the channel attention mechanism.

D. Covariance Matrix Representation

Assuming that the image is $\boldsymbol{A}$, given area $\boldsymbol{R} \in \boldsymbol{A}$, there are H pixels in the region, $\boldsymbol{y}_k$ represent the d-dimensional feature extracted from the kth pixel within $\boldsymbol{R}$, μ denotes the

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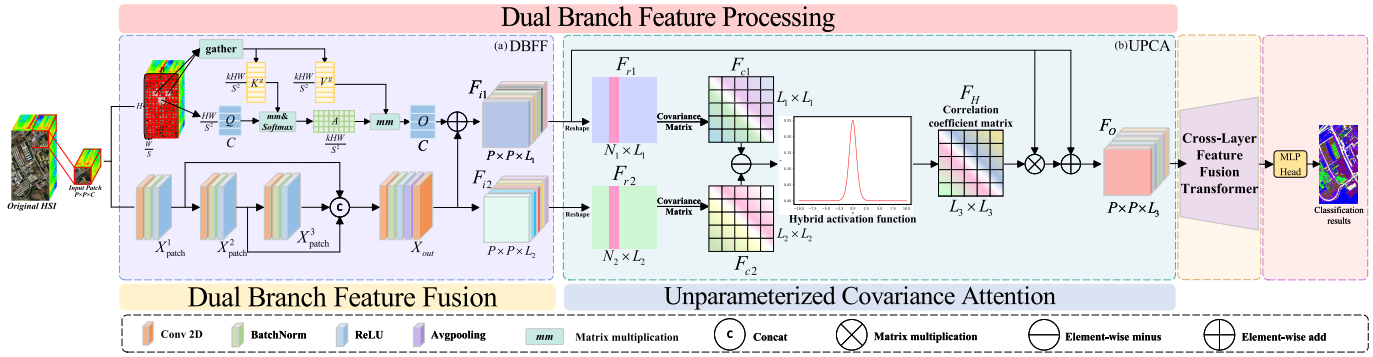


Fig. 1. Proposed CACFTNet network architecture. The network architecture was designed with two main modules: DBFP and CLF2Former. The DBFP module consists of two submodules: (a) DBFF and (b) UPCA.

mean vector of eigenvectors $\boldsymbol{y}_k (k = 1, \ldots, H)$, the covariance matrix representation of area $\boldsymbol{R}$ as shown in the following equation:

$$\boldsymbol{C}_R = \frac{1}{H-1} \sum_{k=1}^{H} (\boldsymbol{y}_k - \mu)(\boldsymbol{y}_k - \mu)^{\mathrm{T}}. \tag{1}$$

The covariance matrix species every off-diagonal value represents the covariance between two identical features.

III. Proposed Method

A. Overall Framework

The CACFTNet framework we designed is shown in Fig. 1 and consists of two main modules: the dual-branch feature processing (DBFP) module and the cross-layer feature fusion transformer (CLF2Former) module. Among them, the DBFP module consists of two submodules: 1) the DBFF module and 2) the UPCA module. Specifically, the input to the network described in this article consists of fixed-size patch data, which initially passes through the DBFP module. Then, spatial–spectral feature information at various scales is extracted from the two branches (local and global), respectively. Next, the features are further processed and fed into the UPCA module to establish correlations between different bands and focus on the important parts of the band information. This input is then used for the CLF2Former module to model long-range dependencies and to represent and learn deep features.

B. Double-Branch Feature Processing

This module primarily focuses on extracting and processing features from two different scales. First, the input features are extracted from two different scales by the global and local feature extraction branches of the DBFF module. Subsequently, the features from the two scales are summed together to facilitate the propagation of multiscale information and enhance the capability of feature expression. Second, the features from various scales are input into the UPCA module to analyze the correlation between different bands, enhance the band information for classification, and enable the network to focus more on the important bands while suppressing redundant ones.

1) Double-Branch Feature Fusion: To extract richer feature information, this article proposes the use of a DBFF module. The structure of DBFF, shown in Fig. 1(a), is proposed in this article. The module consists of a CNN feature extraction module and a transformer-based bi-level routing attention mechanism [47]. These branches are responsible for acquiring spatial and spectral feature information and then integrating the global–local feature information from different scales extracted by them.

a) CNN feature extraction block: To better extract local feature information, we designed a four-layer convolutional approach to extract local features. As depicted in the bottom of Fig. 1(a), each convolutional layer is followed by the corresponding processing. Given an input feature map $\boldsymbol{X} \in \mathbb{R}^{H \times W \times C}$, crop it into multiple $\boldsymbol{X}_{\text{patch}} \in \mathbb{R}^{P \times P \times C}$. First, these HS image patches are passed into 1×1 convolutional blocks to extract spectral feature information. Afterward, the features extracted from the various layers are combined to extract spatial semantic information through convolution with a three-unit kernel. Therefore, combining low-level features with high-level features can be achieved through the use of different receptive fields to capture feature information at various scales in HS images, thus providing a comprehensive feature characterization. The specific parameters of the CNN feature extraction module are provided in Table I. Finally, the number of channels is fixed to an appropriate value using a 1×1 convolution. The details are shown in the following equations:

$$\boldsymbol{X}_{\text{out}} = \text{Conv2d}(\text{AP}(\delta(\text{BN}(\text{Conv2d}(\boldsymbol{X}_{\text{concat}}))))) \tag{2}$$

$$\boldsymbol{X}_{\text{concat}} = \text{Concat}\left(\boldsymbol{X}^1_{\text{patch}}, \boldsymbol{X}^2_{\text{patch}}, \boldsymbol{X}^3_{\text{patch}}\right) \tag{3}$$

$$\boldsymbol{X}^i_{\text{patch}} = \delta(\text{BN}(\text{Conv2d}(\boldsymbol{X}_{\text{patch}}))), \quad (i = 1, 2, 3) \tag{4}$$

where $\boldsymbol{X}_{\text{out}}$ is the output of the entire feature acquisition block; $\boldsymbol{X}^i_{\text{patch}}$ is the output of the ith layer convolution; AP is the AvgPooling average pooling operation; δ is the activation function ReLU; and BN is the batch normalization (BN) operation.

b) Transformer block: HS images are a special type of image in which each pixel contains not only spatial location information but also spectral band information. A bi-level routing attention mechanism based on vision transformer (ViT) [47] can be combined with HS images to achieve more

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TABLE I
FEATURE EXTRACTION MODULE SPECIFIC PARAMETERS

Layer	Operation	Kernel	Padding size	Other
First	Convolution2d	1×1	0	BN,ReLU
Second	Convolution2d	1×1	0	BN,ReLU
Third	Convolution2d	1×1	0	BN,ReLU
Fourth	Convolution2d	3×3	1	BN,ReLU
	AvgPool2d	3×3	1	-
	Convolution2d	1×1	0	-

precise and accurate feature extraction and attention allocation. As shown in the top of Fig. 1(a).

Suppose the input HS image is $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$, H and W represent the height and width of the input image, and C denotes the number of spectral bands. First, the input image $\boldsymbol{X}$ is divided into $S \times S$ different regions, with each region containing (HW/S^2) feature vectors. The input image is reshaped to obtain $\boldsymbol{X}^r \in \mathbb{R}^{S^2\times(HW/S^2)\times C}$. Through linear mapping, the feature vectors in the region are projected to obtain three tensors $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, as shown in the following equation:

$$\boldsymbol{Q} = \boldsymbol{X}^r\boldsymbol{W}^q, \quad \boldsymbol{K}=\boldsymbol{X}^r\boldsymbol{W}^k, \ \boldsymbol{V}=\boldsymbol{X}^r\boldsymbol{W}^v \in \mathbb{R}^{S^2\times\frac{HW}{S^2}\times C} \tag{5}$$

where $\boldsymbol{W}^q$, $\boldsymbol{W}^k$, and $\boldsymbol{W}^v \in \mathbb{R}^{C\times C}$ are the weight values of $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, respectively. $\boldsymbol{X}^r$ is the divided region. In order to obtain correlation information between HS image regions, the average of $\boldsymbol{Q}$ and $\boldsymbol{K}$ within each region is calculated to obtain $\boldsymbol{Q}^r$ and $\boldsymbol{K}^r$ at the regional level. Obtain the correlation matrix $\boldsymbol{A}^r$ by calculating the product of $\boldsymbol{Q}^r$ and $\boldsymbol{K}^r$. The details are shown in the following equation:

$$\boldsymbol{A}^r = \boldsymbol{Q}^r(\boldsymbol{K}^r)^{\mathrm{T}} \in \mathbb{R}^{S^2\times S^2} \tag{6}$$

where the ith and jth rows and columns of the correlation matrix $\boldsymbol{A}$ indicates the degree of relevance between the ith and jth regions, respectively. In order to reduce the computational complexity, only the first k correlation connections of each region are kept to generate the routing index matrix $\boldsymbol{I}^r$. This provides region-level routing information to determine the interactions between each given region and other regions. The details are shown in the following equation:

$$\begin{aligned} &\text{for } i \text{ in range}(S^2):\\ &\quad \text{index} = \text{topkIndex}(\boldsymbol{A}^r[i,:],k)\\ &\quad \boldsymbol{I}^r[i,:] = \text{index} \end{aligned} \tag{7}$$

where $\text{topkIndex}(\boldsymbol{A}^r[i,:],k)$ indicates that the values in the ith row of the correlation matrix are selected and indexed by the first k maximum values. Then, using the routing index matrix $\boldsymbol{I}^r$, we can compute the fine-grained token-to-token attention. For each query token within a region, the system will consider all relevant key-value pairs in the token routing regions. In order to handle the scattered key-value pairs throughout the feature graph, we concentrate on the original keys $(\boldsymbol{K})$ and values $(\boldsymbol{V})$ to derive the aggregated keys $(\boldsymbol{K}^g)$ and aggregated values $(\boldsymbol{V}^g)$. The details are shown in the

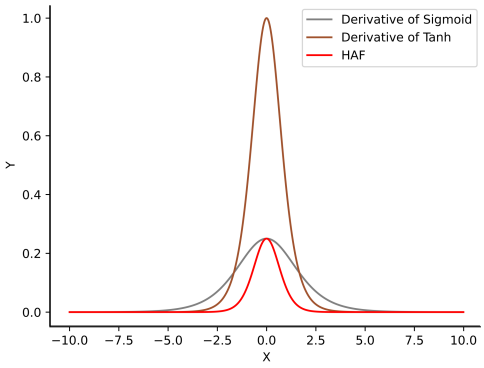


Fig. 2. HAF.

equations:

$$\boldsymbol{K}^g = \text{gather}(\boldsymbol{K},\boldsymbol{I}^r) \in \mathbb{R}^{S^2\times\frac{kHW}{S^2}\times C} \tag{8}$$

$$\boldsymbol{V}^g = \text{gather}(\boldsymbol{V},\boldsymbol{I}^r) \in \mathbb{R}^{S^2\times\frac{kHW}{S^2}\times C}. \tag{9}$$

Then, the output $\boldsymbol{O}$ is obtained based on the attention mechanism of aggregated keys $(\boldsymbol{K}^g)$, aggregated values $(\boldsymbol{V}^g)$, and queries $(\boldsymbol{Q})$. The details as shown in the equations:

$$\boldsymbol{O} = \text{Attention}(\boldsymbol{Q},\boldsymbol{K}^g,\boldsymbol{V}^g) + \text{LCE}(\boldsymbol{V}) \tag{10}$$

$$\text{Attention}(\boldsymbol{Q},\boldsymbol{K}^g,\boldsymbol{V}^g) = \text{Softmax}\left(\frac{\boldsymbol{Q}(\boldsymbol{K}^g)^{\mathrm{T}}}{\sqrt{C}}\right)\boldsymbol{V}^g \tag{11}$$

where $\text{LCE}(\boldsymbol{V})$ denotes the local context enhancement term, parameterized with a depth-separable convolution. $\sqrt{C}$ is used to scale the attention weights so that they are not affected by the number of bands.

In summary, the combination of the bi-level routing attention mechanism with HS images can realize the allocation of attention and feature extraction to different regions in the image, and model the correlation between different areas to capture the key feature information in the HS data.

2) Unparticipated Covariance Attention: In HS image classification, different bands have varying importance for the classification process. There is a specific relationship between bands, as some spectral bands may be redundant and interfere with the classification results. For this reason, this article proposes a UPCA module to explore the correlation between different bands and enhance the information on the bands that are conducive to classification. The UPCA network framework is depicted in Fig. 1(b).

In contrast to most covariance modules, the UPCA proposed in this article fully considers the correlation variations between different bands in the local area, as well as the differing features. After integrating this module into the feature extraction stage, we can obtain more detailed feature information, which will enhance the model's capability for nonlinear modeling.

Taking two features $\boldsymbol{F}_{i1} \in \mathbb{R}^{P\times P\times L_1}$ and $\boldsymbol{F}_{i2} \in \mathbb{R}^{P\times P\times L_2}$ as inputs to the UPCA, where P and L denote the spatial size and number of channels of the input features. Then, $\boldsymbol{F}_{i1}$ and $\boldsymbol{F}_{i2}$ are reshaped and transformed into $\boldsymbol{F}_{r1} \in \mathbb{R}^{N_1\times L_1}$ and $\boldsymbol{F}_{r2} \in \mathbb{R}^{N_2\times L_2}$, respectively. Where N represents the number of pixels in the input features and L denotes that each sample is represented as an L-dimensional vector. Next, the covariance

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matrix is calculated. The details are shown in the following equations:

$$\boldsymbol{F}_{c1} = \frac{1}{N_1 - 1} \sum_{i=1}^{N_1} \left(\boldsymbol{F}_{r1}^{i} - \bar{\boldsymbol{F}}_{r1} \right) \left(\boldsymbol{F}_{r1}^{i} - \bar{\boldsymbol{F}}_{r1} \right)^{\mathrm{T}} \in \mathbb{R}^{L_1 \times L_1} \tag{12}$$

$$\boldsymbol{F}_{c2} = \frac{1}{N_2 - 1} \sum_{i=1}^{N_2} \left(\boldsymbol{F}_{r2}^{i} - \bar{\boldsymbol{F}}_{r2} \right) \left(\boldsymbol{F}_{r2}^{i} - \bar{\boldsymbol{F}}_{r2} \right)^{\mathrm{T}} \in \mathbb{R}^{L_2 \times L_2} \tag{13}$$

where $\boldsymbol{F}_{r1}^{i} \in \mathbb{R}^{L_1}$ and $\boldsymbol{F}_{r2}^{i} \in \mathbb{R}^{L_2}$ denote the ith pixel sample and $\bar{\boldsymbol{F}}_{r1}$ and $\bar{\boldsymbol{F}}_{r2}$ denote the N_1 and N_2 pixel sample mean vectors, respectively. Any element $\boldsymbol{F}_{c1}$ or $\boldsymbol{F}_{c2}$ in the covariance matrix $\boldsymbol{F}_{c1}$ or $\boldsymbol{F}_{c2}$ stands for the relevance between the ith and jth bands within the HS neighborhood, which consolidates the spatial–spectral information. The larger the value in the covariance matrix, the stronger the correlation at the corresponding location.

In order to more clearly compare the variation in correlation and differences in feature information between bands, the difference in correlation between two bands can be quantified by the difference between the covariance matrices $\boldsymbol{F}_{c1}$ and $\boldsymbol{F}_{c2}$. If the difference is low, it indicates a high level of similarity, suggesting the presence of noise or a significant amount of redundant repetitive information at that location. On the contrary, a higher difference indicates lower similarity, suggesting that the location has more distinctive features. Please refer to the following equation for details:

$$\boldsymbol{F}_{\mathrm{diff}} = (\boldsymbol{F}_{c1} - \boldsymbol{F}_{c2}) \in \mathbb{R}^{L_3 \times L_3} \tag{14}$$

where $\boldsymbol{F}_{\mathrm{diff}}$ denotes the difference covariance matrix and $\boldsymbol{F}_{\mathrm{diff}}(i, j)$ represents the difference in relevance between the ith band and the jth band within the HS neighborhood.

In this study, we introduce a new HAF designed to map the differential covariance matrix $\boldsymbol{F}_{\mathrm{diff}}$ to a specific range, emphasizing distinct characteristic differences between various spectral bands. In comparison to the traditional tanh and sigmoid activation functions, the HAF offers a smaller output range, approximately (0, 0.25), as depicted by the red line in Fig. 2. This helps maintain gradient stability, reducing the risk of vanishing or exploding gradients and enhancing the stability of model training. By limiting the output range to a smaller interval, the model's ability to handle outlier inputs can be constrained, thus enhancing the model's resilience and resistance to noise. The specific formula for the HAF activation function is shown in the following equation:

$$\mathrm{HAF}(\boldsymbol{x}) = ((1 - \tanh(\boldsymbol{x})^2)(\mathrm{sig}(\boldsymbol{x})(1 - \mathrm{sig}(\boldsymbol{x})) \tag{15}$$

where $\boldsymbol{x}$ denotes an arbitrary variable and sig denotes the sigmoid activation function. And tanh denotes the tanh activation function.

By utilizing the HAF activation function, we can improve the nonlinear expressive capability of the differential covariance matrix $\boldsymbol{F}_{\mathrm{diff}}$. Building on the earlier feature differencing operation that highlighted differences between two feature maps, HAF further emphasizes these disparities. Specifically, the activation of HAF leads to higher values in regions with significant differences within the differential covariance matrix $\boldsymbol{F}_{\mathrm{diff}}$ after activation, thereby highlighting key areas in the image. This helps to better identify distinct objects and regions and extract crucial information from HS images. It is particularly useful for regions with minor differences, where there may be higher spectral band similarities or noise present. Subsequent activation by HAF assigns lower values to regions with minor differences, effectively suppressing potential noise and compressing redundant information, thereby reducing the data volume of the feature map. The matrix $\boldsymbol{F}_{\mathrm{diff}}$ after HAF activation has shown in the following equation:

$$\boldsymbol{F}_H = \mathrm{HAF}(\boldsymbol{F}_{\mathrm{diff}}) \in \mathbb{R}^{L_3 \times L_3} \tag{16}$$

where $\boldsymbol{F}_H$ represents the matrix of correlation coefficients after activation. Finally, the input features $\boldsymbol{F}_{\mathrm{i1}}$ are multiplied with the correlation coefficient matrix $\boldsymbol{F}_H$ and then summed with $\boldsymbol{F}_{\mathrm{i1}}$ to obtain the final output features $\boldsymbol{F}_O$. This allows for a combination of raw and differential features to further enhance important features

$$\boldsymbol{F}_O = (\boldsymbol{F}_{\mathrm{il}} \otimes \boldsymbol{F}_H) + \boldsymbol{F}_{\mathrm{il}} \tag{17}$$

where $\boldsymbol{F}_O$ denotes the final output of the UPCA module and $\otimes$ denotes matrix multiplication.

C. Cross-Layer Feature Fusion Transformer

Our proposed CLF2Former is based on ViT improvement. First, HS image chunks are grouped after processing. Compared with the traditional ViT, this enables feature interaction and learning to some extent for HS image chunks in the same grouping. Second, the features are further fed into the CAAF module to model the feature information passed between the low-level encoder and the high-level encoder.

1) Groupwise Spectral Embedding (GSE): HS images contain a greater number of spectral channels compared to the blocks of images in a traditional transformer. They are typically sparsely sampled (e.g., every 10 ns) across the electromagnetic spectrum, leading to an almost continuous spectral signature. The spectral information at different locations reflects the absorption properties at various wavelengths, thereby indicating the current physical properties of the material. Therefore, it is crucial to capture the localized detailed absorption or variation of spectral features in HS scenes for accurate and meticulous material classification.

For this purpose, we use the method of previous researchers called the GSE module [21]. This module obtains feature embeddings from the local spectrum of spectral features. As shown in Fig. 3.

In an HS image, each pixel point contains values for multiple spectral channels. So, a set of spectral features (pixel points in a HS image), i.e., $\boldsymbol{y} = \left[y_1, y_2, \ldots, y_m \right] \in \mathbb{R}^{1 \times m}$. Where y_i denotes the value of the ith spectral channel and m denotes the number of spectral channels. The feature embedding $\boldsymbol{b}$ obtained using the classical converter method is shown in the following equation:

$$\boldsymbol{b} = \boldsymbol{\omega} \boldsymbol{y} \tag{18}$$

where $\boldsymbol{\omega} \in \mathbb{R}^{d \times 1}$ represents the weight matrix and the weight matrix $\boldsymbol{\omega}$ can be considered as a linear transformation

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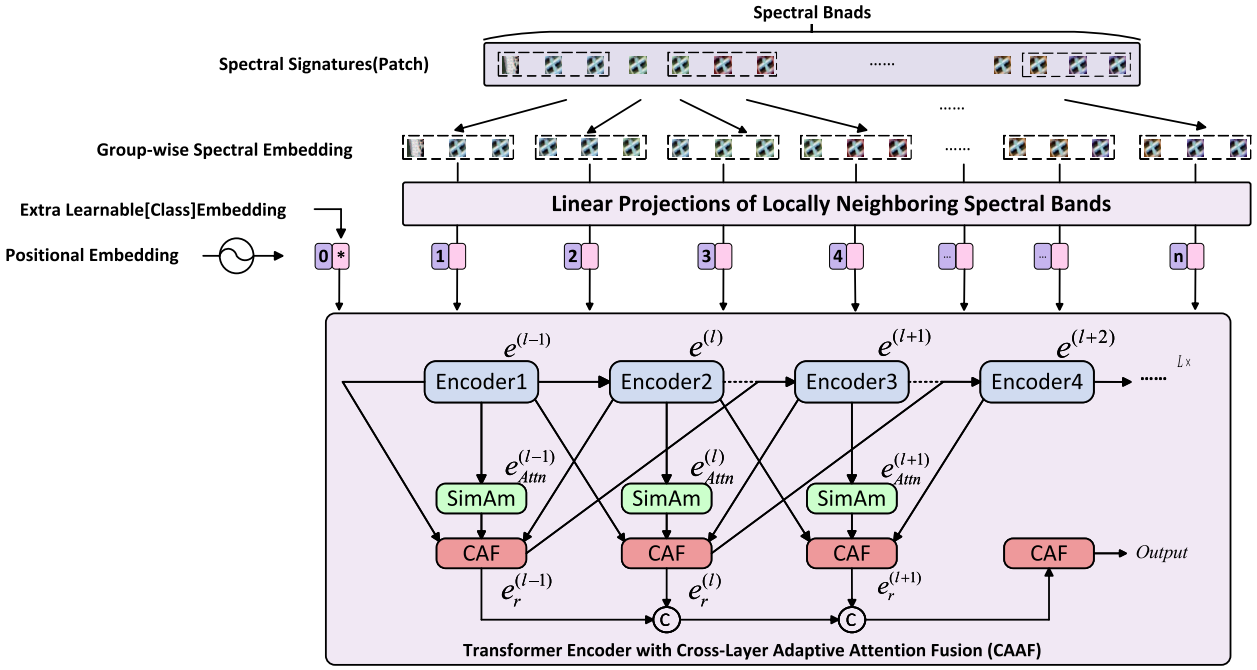


Fig. 3. Proposed CLF2Former structure. It consists of two parts: 1) GSE and 2) CAAF.

operation that maps the input spectral features to the corresponding embedding representations in the feature space. The aim of this linear transformation is to capture the features of different channels in the feature space by processing the spectral channels and generating corresponding feature embedding representations for the linear transformation of the spectral features. $\boldsymbol{b} \in \mathbb{R}^{d\times m}$ denotes the set of feature embeddings of the spectral channels, obtained by multiplying each spectral feature vector $\boldsymbol{y}$ by the weight matrix $\boldsymbol{\omega}$.

In contrast, GSE uses feature embeddings learned in the local spectrum (or adjacent bands) of spectral features. Therefore, the GSE-specific details are shown in the following equation:

$$\boldsymbol{B} = \boldsymbol{W}\boldsymbol{Y} = \boldsymbol{W} f(\boldsymbol{y}) \tag{19}$$

where $\boldsymbol{B} \in \mathbb{R}^{d\times m}$ represents the feature embedding matrix of the GSE. $\boldsymbol{W} \in \mathbb{R}^{d\times n}$ denotes the grouped representation of the variable $\boldsymbol{\omega}$. In the given context, d represents the number of spectral channels in each localized spectrum (or adjacent band), and n denotes the number of adjacent bands. $\boldsymbol{Y} \in \mathbb{R}^{n\times m}$ denotes the grouped variable $\boldsymbol{y}$, where m is the number of pixel points in the image, and n denotes the number of neighboring bands. The spectral feature $\boldsymbol{y}$ is divided into multiple local spectra or neighboring bands, and each local spectrum consists of the spectral channel values of n adjacent bands.

Specifically, the function $f(\cdot)$ represents the grouping of spectral features $\boldsymbol{y}$ based on certain rules and forms a matrix representation $\boldsymbol{Y}$ that captures local details and variations of these features more effectively. Each column $\boldsymbol{y}_q$ in this grouping matrix $\boldsymbol{Y}$ represents a localized group comprising adjacent bands' features. The specific details are illustrated in the following equation:

$$\boldsymbol{Y} = f(\boldsymbol{y}) = [\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q, \ldots, \boldsymbol{y}_m] \tag{20}$$

where $\boldsymbol{y}^q = [\boldsymbol{y}_{q-\lfloor(n/2)\rfloor}, \ldots, \boldsymbol{y}_q, \ldots, \boldsymbol{y}_{q+\lfloor(n/2)\rfloor}]^{\mathrm{T}} \in \mathbb{R}^{n\times 1}$, which are localized groups $\boldsymbol{y}^q$ that are extracted from the original spectral features $\boldsymbol{y}$ by means of the overlapping grouping operation function $f(\boldsymbol{y})$. And $\lfloor\cdot\rfloor$ denotes the round operation.

2) Cross-Layer Adaptive Attention Fusion: Shallow information can provide fundamental visual features such as color, texture, and shape, which typically have high spatial resolution and can directly represent the appearance of an object. Second, in-depth information can offer more abstract and high-level semantic information. To fully utilize both low-level and high-level features simultaneously. As the number of encoder layers increases, advanced encoder features may lose shallow information, the expressive ability of low-level features may be much less than that of high-level features, and their detailed information may be insufficient. Therefore, we employ cross-layer attentional connectivity to integrate crucial information across different layers. This approach focuses on the significant information in both the low-level and high-level encoder features, enhancing the expressive capacity of the features and reducing redundant information. This approach allows us to obtain more comprehensive multilevel information, making it better suited for HS images to achieve precise and detailed classification. For this reason, we designed a CAFF module (as shown in Fig. 3).

Cross-layer adaptive fusion (CAF): $e^{(l-1)} \in \mathbb{R}^{1\times d_e}$ represents the $(l-1)$th layer encoder output; $e^{(l)}_{\mathrm{Attn}} \in \mathbb{R}^{1\times d_e}$ represents the (l)th layer encoder output after attention; $e^{(l+1)} \in \mathbb{R}^{1\times d_e}$ represents the $(l+1)$th layer encoder output. The structure is shown in Fig. 4. The output of CAF is shown in the following equation:

$$e^{(l)}_r \xleftarrow{\mathrm{Conv}} \ddot{\boldsymbol{\omega}} \begin{bmatrix} e^{(l-1)} \\ e^{(l)}_{\mathrm{Attn}} \\ e^{(l+1)} \end{bmatrix} \tag{21}$$

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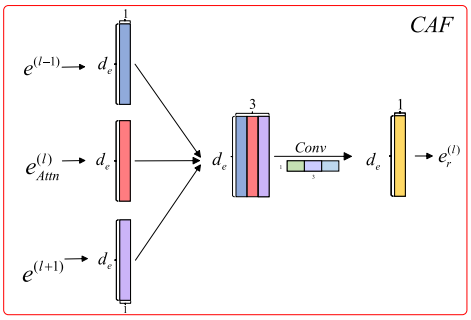


Fig. 4. Diagram of the CAF module in the proposed CACFTNet.

where $e_r^{(l)}$ denotes the result of the (l)th layer of fusion, and $\ddot{\boldsymbol{\omega}} \in \mathbb{R}^{1\times 3}$ are network parameters that can be learned and used for adaptive fusion.

During the process of performing CAF fusion, we fused two original encoder features with a pass-attention encoder. This is done for two main important reasons. First, there will be an obvious semantic feature gap between the shallow and deep encoders extracting two different features, respectively. In order to prevent the feature information from spanning too much and the loss of information, they are allowed to perform feature fusion with the encoder that passes the attention, which serves to supplement the gap information. Second, two different levels of feature information can be retained and feature information of deeper importance can be added to effectively eliminate redundancy between different layers.

However, in previous work [21], the fusion process skips an encoder's feature information, which may make the final fusion not sufficient and lead to degradation of network performance. Instead, this is fully taken into account in our work. We introduce a parameter-free attention mechanism-SimAm [30], which generates a 3-D attention weight, and then fully fuses it with the original feature map to highlight the more important parts of the feature map without adding extra parameters to the network, which is a lightweight parameter-free attention. Fusing the feature information after attention with shallow and deep features can effectively model the dependency relationship between features, enhance the discriminative ability of features, and improve the effect of fusion. For example, in Fig. 3, when the feature information of encoder 2 passes through attention, attention can adaptively assign the attention weights according to the importance of its different features to highlight the more critical parts, and filter out the useless feature information, retain only the feature information that contributes to the classification, and then finally fuse it with encoders 1 and 3.

D. Cross Entropy Loss Function

In this work, we choose the cross-entropy loss function. The cross-entropy loss function is a loss commonly used for multiclassification, which can effectively measure the gap between the predicted results and the real labels, making the model estimate the classification probability more accurately and minimize the difference between the actual classification results and the real labels. The cross-entropy loss function is a loss commonly used for multiclassification, which can effectively measure the gap between the predicted results and the true labels, allowing the model to more accurately estimate the classification probability and minimize the difference between the actual classification results and the true labels

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(p_{ij}) \tag{22}$$

where N is the number of samples, C is the number of categories, y_{ij} is the true label of the sample i belonging to the category j, and p_{ij} denotes the probability that the model predicts that the sample i belongs to the category j.

Compared with other loss functions, the gradient of the cross-entropy loss function is simpler, which helps our algorithm converge faster to improve the training effect. What's more, the HS image classification task, usually encounters the situation of category imbalance, i.e., the number of samples of different categories varies greatly, which may misidentify a small number of samples as a large number of samples. This loss can effectively reduce misclassification and improve the classification performance of the model.

IV. Experiments

In this section, three well-known HS datasets are first introduced; then the details of our network implementation as well as extensive ablation experiments on the proposed modules and innovative HAFs are presented; and finally, comparative experiments are conducted to further evaluate the HS image classification performance of CACFTNet.

A. Datasets

To further validate the classification performance of CACFTNet, we selected three datasets that are well-known in the HS field. They are Indian Pines (IP), Pavia University (PU), and Houston2013 (HU2013) datasets.

1) IP Dataset: The IP dataset was collected by the Airborne/Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 in northwestern IN, USA. It consists of 145 × 145 pixels divided into 16 vegetation classes with 10 366 samples, and the ground sampling distance (GSD) is 20 m. It contains 220 spectral bands in the wavelength range of 0.4–2.5 μm, and 200 spectral bands are retained for the study after removing 20 unwanted bands. The range of bands is divided into three phases: 1–103, 109–149, and 164–219.

2) PU Dataset: The PU dataset was acquired by the Reflectance Optical System Imaging Spectrometer (ROSIS) sensor during flight operations over the University of Pavia, Northern Italy. The dataset is composed of 610 × 340 pixels, with a total of 43 923 samples in nine land types, containing 115 spectral bands in the wavelength range of 0.43–0.86 μm, removing 12 noisy bands, and retaining 103 spectral bands as objects of study.

3) HU2013 Dataset: The HU2013 dataset centers on the agricultural area of the University of Houston and the surrounding area in the United States and was obtained by ITRES CASI-1500. It is composed of 349 × 1905 pixels with 15 land types and a total of 15 029 samples containing 144 spectral bands in the wavelength range of 0.38–1.05 μm, and the GSD is 2.5 m.

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TABLE II
LAND-COVER CLASSES OF THE IP DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
C.1	Corn Notill	50	1384
C.2	Corn Mintill	50	784
C.3	Corn	50	184
C.4	Grass Pasture	50	447
C.5	Grass Trees	50	697
C.6	Hay Windrowed	50	439
C.7	Soybean Notill	50	918
C.8	Soybean Mintill	50	2418
C.9	Soybean Clean	50	564
C.10	Wheat	50	162
C.11	Woods	50	1244
C.12	Buildings Grass Trees Drives	50	330
C.13	Stone Steel Towers	50	45
C.14	Alfalfa	15	39
C.15	Grass Pasture Mowed	15	11
C.16	Oats	15	5
	Total	695	9671

TABLE III
LAND-COVER CLASSES OF THE PU DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
C.1	Asphalt	548	6304
C.2	Meadows	540	18146
C.3	Gravel	392	1815
C.4	Trees	524	2912
C.5	Metal Shects	265	1113
C.6	Bare Soil	532	4572
C.7	Bitumen	375	981
C.8	Bricks	514	3364
C.9	Shadows	231	795
	Total	3921	40002

TABLE IV
LAND-COVER CLASSES OF THE HU2013 DATASET, AND THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class No.	Class Name	Training	Testing
C.1	Healthy Grass	198	1053
C.2	Stressed Grass	190	1064
C.3	Synthetic Grass	192	505
C.4	Tree	188	1056
C.5	Soil	186	1056
C.6	Water	182	143
C.7	Residentiał	196	1072
C.8	Commercial	191	1053
C.9	Road	193	1059
C.10	Highway	191	1036
C.11	Railway	181	1054
C.12	Parking Lot1	192	1041
C.13	Parking Lot2	184	285
C.14	Tennis Court	181	247
C.15	Running Track	187	473
	Total	2832	12197

B. Experimental Setting

1) Evaluation Metrics: To better analyze the superiority of our method over other methods in terms of classification performance, we use three common evaluation metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). As each metric gets higher, the better the model's classification results.

2) Comparative Approaches: To evaluate the effectiveness of the proposed CACFTNet, we conducted comparative experiments with traditional methods, CNN-based methods, and transformer-based methods. This includes SVM [5], 1-D CNN [13], 2-D CNN [10], spatial–spectral residual network (SSRN) [48], ViT [11], end-to-end IFormer [43], GAHT [40], SpectralFormer [21], multivisual fusion network (MVAHN) [23], morphological transformer (morphFormer) [22], and our proposed CACFTNet.

1) For 1-D CNN, it consists of the following components: 1-D convolution, BN layer, rectified linear unit (ReLU) layer, and softmax activation function.
2) For 2-D CNN, there are three 2-D convolution blocks. Each block has convolution kernels of sizes $3 \times 3 \times 32$, $3 \times 3 \times 32$, and $1 \times 1 \times 128$, respectively. Each convolution block consists of a 2-D convolution layer, a BN layer, and a max pooling layer.
3) For SSRN, the input data consists of original data cubes, and it utilizes two residual blocks to extract the spectral–spatial features. The spectral feature extraction part consists of six convolution layers with kernel sizes of $1 \times 1 \times 7$ and $1 \times 1 \times 97$. The spatial feature extraction part consists of five convolution layers, one average pooling layer, and one fully connected layer. Each convolution layer has a kernel size of $3 \times 3 \times 128$.
4) For ViT, the number of transformer encoders is set to 5, and the length of token embeddings is 64. For the specific network structure, please refer to reference [11].
5) For IFormer, the HS image blocks are processed sequentially through 1-D convolution and BN operations to extract spectral features. The Ghost module is then utilized to fully exploit the intrinsic information in the HS image features. Next, the IFormer encoder is employed to learn the combination of low-frequency and high-frequency features in the feature maps. Finally, the learned features are applied to the classification task through pooling layers and linear layers.
6) For GAHT, it is mainly divided into three consecutive stages. Each stage consists of a grouped pixel embedding module and several transformer blocks. After the third stage, spatial information is compressed through global average pooling, and classification is performed using fully connected layers.
7) For SpectralFormer, the input tokens are passed through five transformer encoder blocks. Each encoder block has four attention heads.
8) For MVAHN, first, local features are extracted through a spectral–spatial convolution module. Then, these features are fed into ViT and processed by convolutional embedding transformer encoders and graph convolution modules to enhance the capability of local feature extraction and improve attention to the similarity relationships between pixels. Finally, a linear classifier is applied to obtain the classification results.

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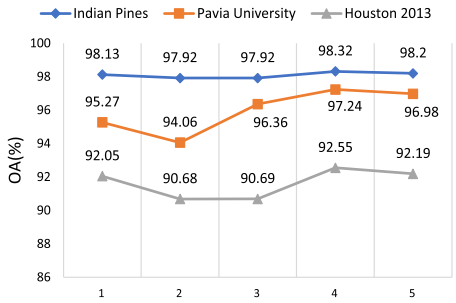


Fig. 5. Comparison of OA accuracy with the different number of attention heads.

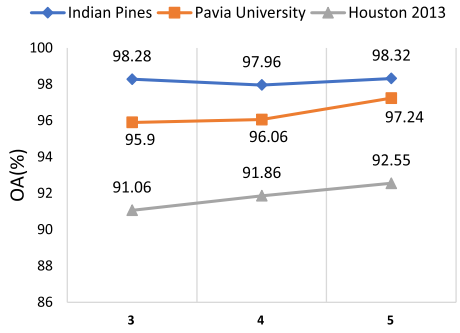


Fig. 6. Comparison of OA accuracy at different transformer encoder depths.

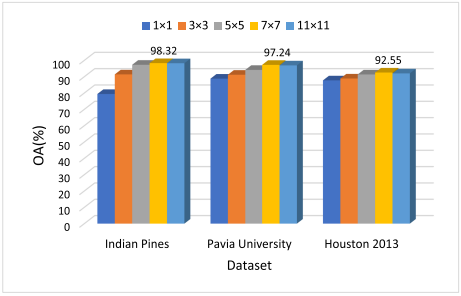


Fig. 7. Comparison of OA accuracy for three datasets with different patch sizes.

9) For morphFormer, the discriminative features in the HS image are extracted using 3-D convolution blocks and heterogeneous 2-D convolution blocks. BN layers and ReLU activation layers are used after each block. Then, the features are reshaped into patch tokens and fed into the encoders of ViT, with eight attention heads. Finally, the classification result is obtained by the classifier head from the classification tokens.

3) Implementation Details: In the experiments, we randomly select a small number of samples from each category to constitute the training dataset and use the remaining samples for testing. The details are shown in Tables II–IV. We designed CACFTNet based on a PyTorch implementation using a workstation with an NVIDIA TITAN RTX (24 GB) GPU. The Adam optimizer is used, with a batch size of 64, each patch size set to 7×7, an initialized learning rate value of 5×10^{-4}, and a weight decay value set to 5×10^{-3}. In addition, in the transformer encoder part, the encoder depth is 5, and the number of attention heads is 4. It is worth noting that during the experiments, the feature extraction ability of the network may show differences when dealing with different datasets.

C. Parameter Sensitivity Analysis

For the purpose of assessing the function of the proposed CACFTNet model more comprehensively, we have investigated several key hyperparameters in the network. The main focus is on the depth of the transformer encoder, the number of heads in the attention, the size of the HS image patch, and the reasonable settings of these parameters affect the performance of our model to some extent. We adopt the control variable method to fix the other hyperparameters in the model and study and analyze the hyperparameters proposed above.

1) Transformer Hyperparameter Analysis: First, in order to investigate the impact of the number of attention heads on the model, we fixed the other parameters and adjusted the number of attention heads from 1 to 5 to observe the effect of the optimal number of attention heads on the model. Second, as shown in Fig. 5, it is evident that the optimal number of attention heads is 4. Therefore, we set the number of attention heads to 4.

Second, the depth of the transformer encoder is crucial for the classification results of our network. Therefore, with the number of attention heads set to 4 and other parameters fixed, we adjust the encoder depth from 3 to 5 and compare the impact of the three different encoder depths on the network accuracy one by one. As shown in Fig. 6, it is obvious that when the depth of the transformer encoder is 5, the model achieves optimal classification performance.

2) Patch Size Impact Analysis: In order to further explore the impact of patch size on the model, we will select one optimal result from five patch sizes, namely, 1×1, 3×3, 5×5, 7×7, and 11×11. Fig. 7 gives the histogram results for different patch sizes on the three datasets. It can be clearly seen that the global accuracy of the 7×7 patch size is the highest on all three datasets, therefore, we choose the 7×7 patch size as the optimal size. As the patch size increases, the OA accuracy shows a low-high-low trend, mainly because larger patch sizes contain richer information, but too large a patch size also brings more redundant information.

D. Ablation Study

1) Main Module Ablation Experiments: In order to make the modules in our proposed CACFTNet model more convincing, we conduct ablation experiments on three datasets with different combinations of modules. Our model can be regarded as consisting of three components: the DBFF module, the UPCA module, and the CAAF module. As shown in Table V, eight combinations of the three components are demonstrated. The effectiveness of the modules is mainly analyzed in terms of three different evaluation indexes: OA, AA, and Kappas.

It is evident that when all three modules are used simultaneously, there is a significant improvement in model accuracy. Compared to not using any of the three modules, the OA metrics for the three datasets have improved by 16.56%, 6.17%, and 4.54%, respectively. Taking the IP dataset as an example, using any single module compared to not using any

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TABLE V

RESULTS OF ABLATION STUDIES OF THE PROPOSED MODULE ON DIFFERENT DATASETS. THE BOLD ONE IS THE OPTIMAL RESULT

Datasets	DBFF	UPCA	CAAF	OA(%)	AA(%)	Kappa
IP	✗	✗	✗	81.76	87.81	0.7919
	✓	✗	✗	97.86	98.91	0.9754
	✗	✗	✓	84.43	88.78	0.8225
	✗	✓	✗	89.00	92.96	0.8743
	✓	✓	✗	97.79	98.79	0.9746
	✗	✓	✓	85.36	89.62	0.8320
	✓	✗	✓	91.16	95.52	0.8991
	✓	✓	✓	**98.32**	**99.11**	**0.9808**
PU	✗	✗	✗	91.07	90.20	0.8805
	✓	✗	✗	94.95	94.55	0.9320
	✗	✗	✓	88.99	86.26	0.8493
	✗	✓	✗	92.36	89.56	0.8955
	✓	✓	✗	96.73	**96.50**	0.9559
	✗	✓	✓	92.69	91.03	0.9012
	✓	✗	✓	95.29	95.10	0.9362
	✓	✓	✓	**97.24**	96.25	**0.9627**
HU2013	✗	✗	✗	88.01	88.91	0.8699
	✓	✗	✗	92.14	92.71	0.9146
	✗	✗	✓	85.86	86.93	0.8464
	✗	✓	✗	85.38	86.43	0.8413
	✓	✓	✗	91.29	92.51	0.9055
	✗	✓	✓	86.68	87.63	0.8554
	✓	✗	✓	90.12	91.21	0.8927
	✓	✓	✓	**92.55**	**93.13**	**0.9191**

module led to improvements in the OA by 16.1%, 2.67%, and 7.24%, respectively. It is worth noting that in all three datasets, an increase in accuracy is observed when any two modules are already present and the remaining modules are added. This phenomenon clearly indicates that the three modules we have designed are effective for all three datasets.

2) Activation Function Ablation Experiment: We tested several more commonly used activation functions: ReLU, Gaussian error linear units (GeLU), sigmoid, sigmoid derivative, tanh, and tanh derivative to verify that our improved HAF makes our network better stabilized. As shown in Table VI, HAT performs significantly better than the other six activation functions in the three evaluation metrics.

3) CAAF Module Ablation Experiments: In order to verify that the proposed cross-layer adaptive connectivity structure and attentional feature fusion are effective, we have done three sets of ablation experiments on each of the three datasets as follows. 1) In the presence of the UPCA module and the DBFF module, only the structure of the five-encoder serial is used without any fusion strategy. Expressed as (without CAAF). 2) Using the CAAF module, provided that the first two modules are present, but removing the attention part of it. Expressed as (without attention). 3) On the premise that all three modules, UPCA, DBFF, and CAAF, are fully present. Expressed as GeLU [(ours(all)].

This can be clearly seen in Table VII, using the IP dataset as an example (the situation for the remaining two datasets is the same as for the IP dataset). First, all three evaluation metrics are steadily increasing for all three structures from top to bottom. The accuracy of removing only the CAAF attention part (without attention) is obviously higher than

TABLE VI

ACCURACY RESULTS FOR DIFFERENT ACTIVATION FUNCTIONS ON EACH OF THE THREE DATASETS. THE BOLD ONE IS THE OPTIMAL RESULT

Datasets	Function	OA(%)	AA(%)	Kappa
IP	ReLU	97.48	98.72	0.9710
	GeLU	97.30	98.75	0.9690
	Sigmoid	98.13	98.96	0.9785
	Sigmoid Derivative	97.95	98.70	0.9765
	Tanh	97.20	98.43	0.9678
	Tanh Derivative	97.87	98.91	0.9756
	HAT(ours)	**98.32**	**99.11**	**0.9808**
PU	ReLU	96.66	96.35	0.9550
	GeLU	95.58	96.76	0.9409
	Sigmoid	96.14	95.90	0.9479
	Sigmoid Derivative	96.90	95.97	0.9580
	Tanh	95.55	94.59	0.9395
	Tanh Derivative	96.38	**97.04**	0.9513
	HAT(ours)	**97.24**	96.25	**0.9627**
HU	ReLU	92.16	93.12	0.9149
	GeLU	91.63	92.75	0.9091
	Sigmoid	91.60	92.07	0.9088
	Sigmoid Derivative	92.02	93.07	0.9134
	Tanh	90.13	91.79	0.8929
	Tanh Derivative	91.01	91.96	0.9025
	HAT(ours)	**92.55**	**93.13**	**0.9191**

TABLE VII

ABLATION ANALYSIS OF THE PROPOSED STRUCTURE ON THREE DATASETS. THE BOLD ONE IS THE OPTIMAL RESULT

Datasets	structure	OA(%)	AA(%)	Kappa
IP	Without CAAF	97.62	98.49	0.9727
	Without Attention	97.85	98.67	0.9753
	Ours(All)	**98.32**	**99.11**	**0.9808**
PU	Without CAAF	94.63	95.62	0.9281
	Without Attention	94.87	95.21	0.9307
	Ours(All)	**97.24**	**96.25**	**0.9627**
HU2013	Without CAAF	90.74	92.04	0.8994
	Without Attention	91.92	92.62	0.9123
	Ours(All)	**92.55**	**93.13**	**0.9191**

that of not adding any fusion strategy (without CAAF), which indicates that our designed cross-layer fusion adaptive connection structure can effectively fuse the important features of the shallow and deep layers between different encoders. In addition, it can be noticed that the accuracy of the CAAF structure with complete presence [ours(all)] is 0.4%–0.7% points higher than the accuracy of removing only the CAAF attention part (without attention). This also illustrates on the basis of cross-layer adaptive connectivity, feature fusion with added attention is able to fuse the more important feature parts with shallow and deep features, which makes the fusion process more adequate and reduces the redundancy of features.

E. Comparison Experiments

1) Quantitative Analysis: We conducted extensive experiments on three datasets, namely, the IP dataset, the PU dataset, and the HU2013 dataset, to analyze and compare the performance of the evaluation metrics OA, AA, and Kappa. The comparative results are presented in Tables VIII–X. Based on these findings, several conclusions can be drawn. First, traditional methods such as SVM, which rely solely on manually crafted features, struggle to accurately classify pixels. Second, classical CNN-based approaches (1D-CNN, 2D-CNN, and SSRN) demonstrate superior local modeling

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TABLE VIII

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE IP DATASET

Class No.	Classifier	CNNs			Transformers						
	SVM [5]	1D-CNN [13]	2D-CNN [10]	SSRN [48]	ViT [11]	iFormer [43]	GAHT [40]	SpectralFormer [21]	MVAHN [23]	morphFormer [22]	**CACFTNet(ours)**
1	67.34	58.67	70.95	93.85	54.04	71.38	86.19	59.68	91.61	91.91	**95.88**
2	67.86	52.93	94.00	98.21	62.62	87.24	94.89	80.48	96.68	97.32	**99.74**
3	93.48	80.43	98.91	**100.00**	82.60	98.36	94.56	85.32	**100.00**	**100.00**	**100.00**
4	94.63	91.94	83.44	95.97	86.80	96.42	93.95	93.06	95.97	**99.55**	97.76
5	88.52	78.33	**100.00**	99.42	81.49	99.71	**100.00**	87.23	**100.00**	99.85	**100.00**
6	94.76	95.89	95.44	99.77	91.57	**100.00**	99.77	95.89	**100.00**	99.77	**100.00**
7	73.86	68.62	88.67	95.75	78.32	81.37	86.49	78.64	93.46	93.89	**98.80**
8	52.07	64.10	45.24	91.56	70.22	76.88	86.10	73.94	96.23	94.95	**97.89**
9	72.70	64.00	65.60	94.85	43.43	77.12	89.36	78.72	93.26	94.32	**96.98**
10	98.77	99.38	**100.00**	**100.00**	98.76	**100.00**	**100.00**	99.38	**100.00**	**100.00**	**100.00**
11	86.17	88.82	86.01	98.79	94.05	93.81	95.65	93.72	**99.11**	97.50	98.95
12	71.82	64.84	98.78	98.78	62.72	97.57	89.69	78.48	**100.00**	**100.00**	99.69
13	95.56	97.77	**100.00**	**100.00**	95.55	**100.00**	**100.00**	97.77	**100.00**	**100.00**	**100.00**
14	82.05	76.92	69.23	**100.00**	84.61	94.87	92.30	69.23	**100.00**	97.17	**100.00**
15	90.91	90.90	**100.00**	**100.00**	81.81	**100.00**	**100.00**	90.90	**100.00**	**100.00**	**100.00**
16	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	72.36	70.96	75.63	95.73	72.77	84.87	90.90	79.46	96.28	96.04	**98.32**
AA(%)	83.16	79.60	87.27	97.94	79.29	92.17	94.31	85.16	97.90	97.27	**99.11**
Kappa	0.6888	0.6699	0.7277	0.9511	0.6883	0.8277	0.8959	0.7663	0.9573	0.9546	**0.9808**

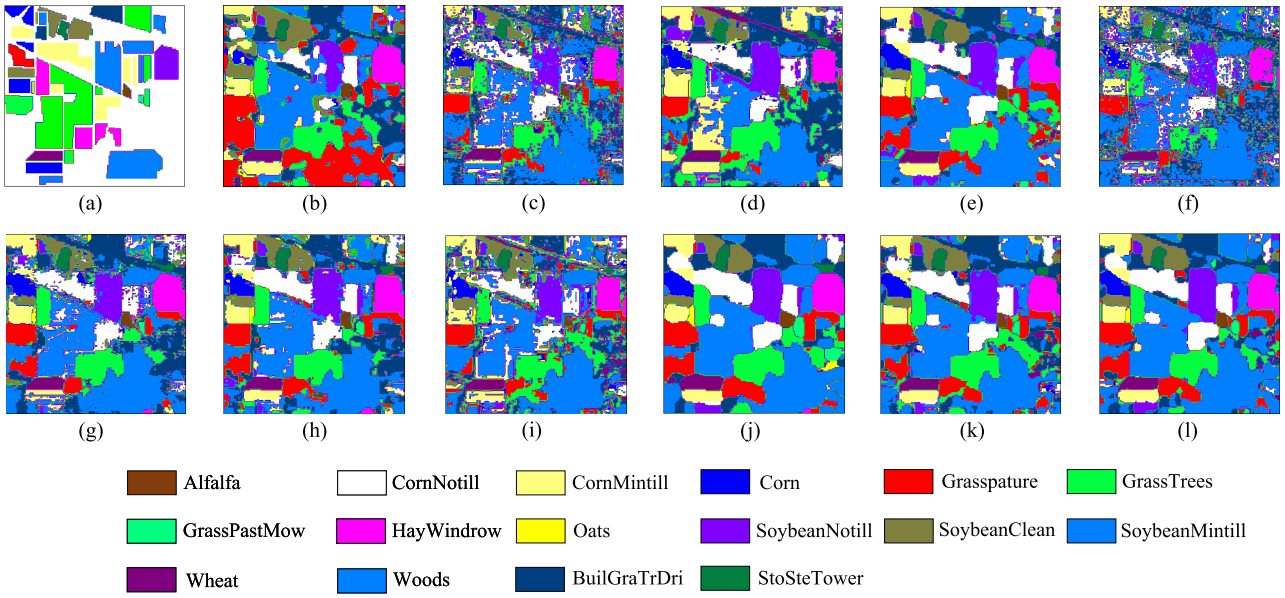


Fig. 8. Classification visualization maps of all methods on the IP dataset. (a) Real ground feature map. (b)–(l) Classification map of SVM, 1D-CNN, 2D-CNN, SSRN, ViT, iFormer, GAHT, SpectralFormer, MVAHN, morphFormer, and CACFTNet (ours), respectively.

TABLE IX

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE PU DATASET

Class No.	Classifier	CNNs			Transformers						
	SVM [5]	1D-CNN [13]	2D-CNN [10]	SSRN [48]	ViT [11]	iFormer [43]	GAHT [40]	SpectralFormer [21]	MVAHN [23]	morphFormer [22]	**CACFTNet(ours)**
1	74.22	71.70	84.77	97.43	79.52	**97.81**	94.35	77.20	94.84	90.21	94.97
2	52.79	69.62	93.85	93.03	74.40	97.36	96.86	96.15	98.83	98.46	**99.36**
3	65.45	52.87	61.76	**85.01**	67.60	74.54	71.62	75.31	84.24	81.59	83.69
4	97.42	91.86	**99.00**	98.00	94.74	97.87	95.74	97.35	95.98	83.48	96.70
5	99.46	96.04	99.92	97.03	99.37	98.38	98.92	99.37	92.90	**100.00**	98.20
6	93.48	**96.69**	57.72	82.61	83.59	62.22	88.62	77.05	92.97	70.42	94.75
7	87.87	76.65	96.94	**100.00**	89.80	79.40	82.05	93.27	99.49	95.92	**100.00**
8	89.39	93.87	98.72	97.44	84.80	97.71	97.17	93.90	97.47	99.58	**99.70**
9	**99.87**	82.64	98.11	99.49	97.73	98.74	94.84	89.18	98.74	98.36	98.86
OA(%)	70.82	76.88	87.95	93.32	79.85	92.07	93.98	89.82	96.40	92.18	**97.24**
AA(%)	84.44	85.78	87.86	94.45	85.73	89.34	91.14	88.76	95.05	90.90	**96.25**
Kappa	0.6423	0.7068	0.8363	0.9101	0.7407	0.8912	0.9188	0.8625	0.9513	0.8929	**0.9627**

capabilities due to convolution operations, often achieving better results compared to traditional methods. However, it should be noted that CNNs have inherent limitations in capturing global features accurately. Third, it was surprising to observe that the performance of the ViT was comparable to 1D-CNN in the context of HS image classification. This

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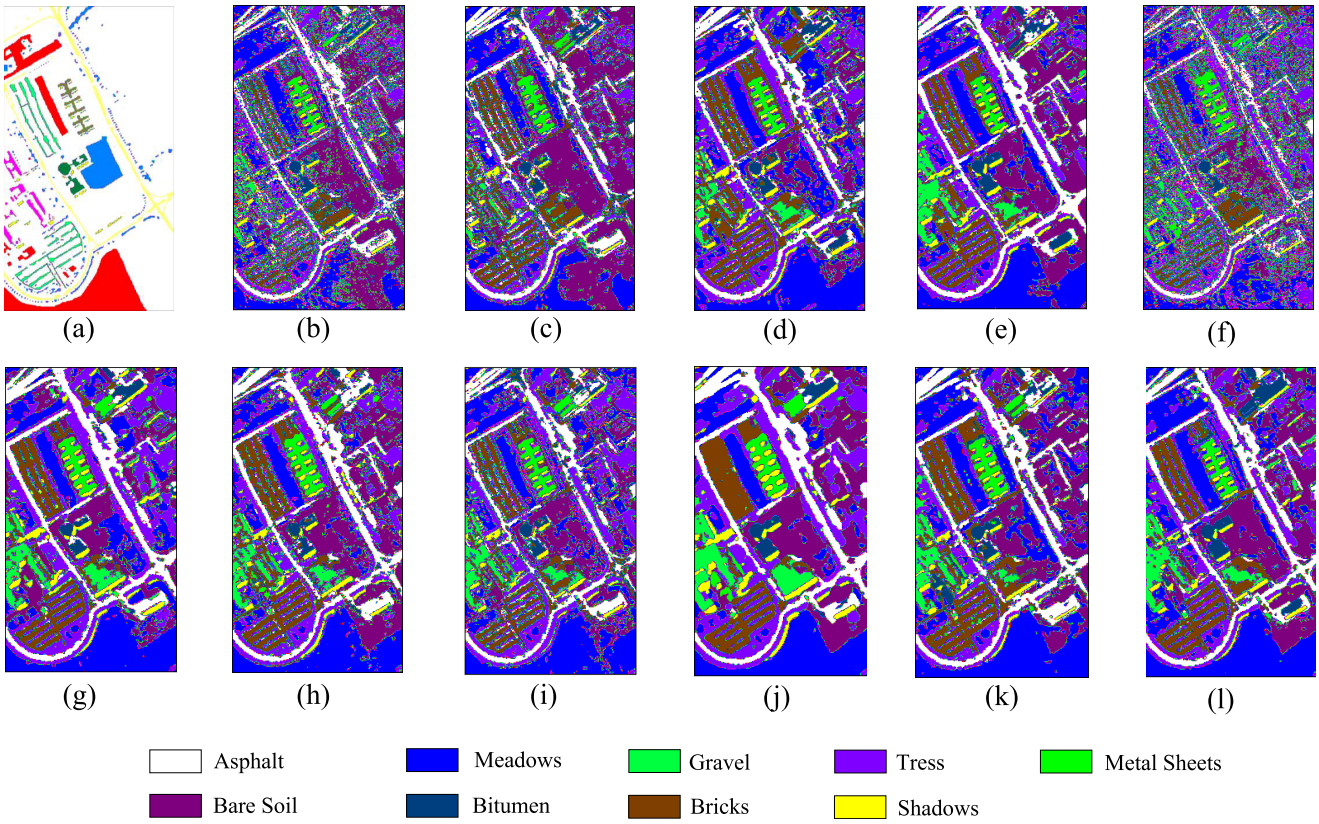


Fig. 9. Classification visualization maps of all methods on the PU dataset. (a) Real ground feature map. (b)–(l) Classification map of SVM, 1D-CNN, 2D-CNN, SSRN, ViT, iFormer, GAHT, SpectralFormer, MVAHN, morphFormer, and CACFTNet (ours), respectively.

TABLE X
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE HU2013 DATASET

Class No.	Classifier	CNNs			Transformers						
	SVM [5]	1D-CNN [13]	2D-CNN [10]	SSRN [48]	ViT [11]	iFormer [43]	GAHT [40]	SpectralFormer [21]	MVAHN [23]	morphFormer [22]	**CACFTNet(ours)**
1	83.00	83.09	83.76	87.55	83.38	82.24	83.09	**86.22**	79.39	83.00	84.42
2	98.40	96.05	94.64	98.49	94.92	**100.00**	97.74	98.96	99.81	99.43	99.71
3	99.60	89.50	93.46	**99.80**	99.60	89.70	90.69	89.90	99.60	**99.80**	**99.80**
4	98.48	97.91	**100.00**	99.90	98.86	99.90	99.81	97.34	99.05	97.82	99.62
5	97.82	96.02	96.02	97.82	97.72	**100.00**	99.90	99.71	99.81	99.81	**100.00**
6	90.91	86.01	90.20	96.50	95.10	90.90	94.40	96.50	**100.00**	95.80	95.80
7	90.39	91.41	81.62	92.91	80.78	93.56	**93.84**	84.32	69.30	88.52	89.64
8	40.46	74.54	48.33	92.21	52.42	**88.60**	74.54	79.86	63.91	83.00	88.12
9	41.93	77.33	**90.46**	88.57	69.78	85.93	89.14	76.10	82.43	87.15	75.54
10	62.64	45.84	61.29	66.79	84.94	59.45	77.41	79.44	75.57	80.01	**98.06**
11	75.43	74.09	82.73	98.95	68.97	97.24	94.11	87.19	**100.00**	95.92	99.62
12	60.04	64.07	73.19	82.51	48.31	82.99	86.07	78.57	**93.75**	58.59	84.53
13	49.47	77.19	83.50	86.66	65.26	90.87	**91.57**	73.68	81.75	91.22	82.10
14	98.79	92.30	99.59	**100.00**	99.59	96.76	99.59	98.78	**100.00**	90.28	**100.00**
15	97.46	97.88	74.63	**100.00**	98.94	99.78	95.34	99.78	99.78	95.77	**100.00**
OA(%)	76.91	81.48	82.05	91.55	80.10	89.74	90.19	87.49	87.71	90.78	**92.55**
AA(%)	78.99	82.89	83.57	92.58	82.58	90.53	91.15	88.43	89.61	91.54	**93.13**
Kappa	0.7494	0.7989	0.8052	0.9082	0.7846	0.8885	0.8934	0.8642	0.8666	0.8999	**0.9191**

suggests that both methods have similar effectiveness in this task. The reason for this parity may lie in the fact that 1D-CNN focuses solely on spectral information, and both ViT and 1D-CNN overlook the influence of neighboring pixels, leading to a significant decrease in classification accuracy. Fourth, while the other five methods (iFormer, GAHT, SpectralFormer, MVAHN, and morphFormer) demonstrate significant potential in establishing global contextual dependencies, relying solely on a transformer-based classification framework does not yield satisfactory results, as seen in the case of SpectralFormer. Conversely, combining CNN and transformer in the classification algorithm has shown promising results, as demonstrated by iFormer, GAHT, MVAHN, morphFormer, and the proposed CACFTNet network. However, these five methods still lag behind in terms of the OA metric on the HU2013 dataset, with respective deficits of 2.81%, 2.36%, 5.06%, 4.84%, and 1.77%.

It is well known that each dataset presents unique challenges in the identification of various shape regions. Therefore, we conducted a detailed analysis of the challenging

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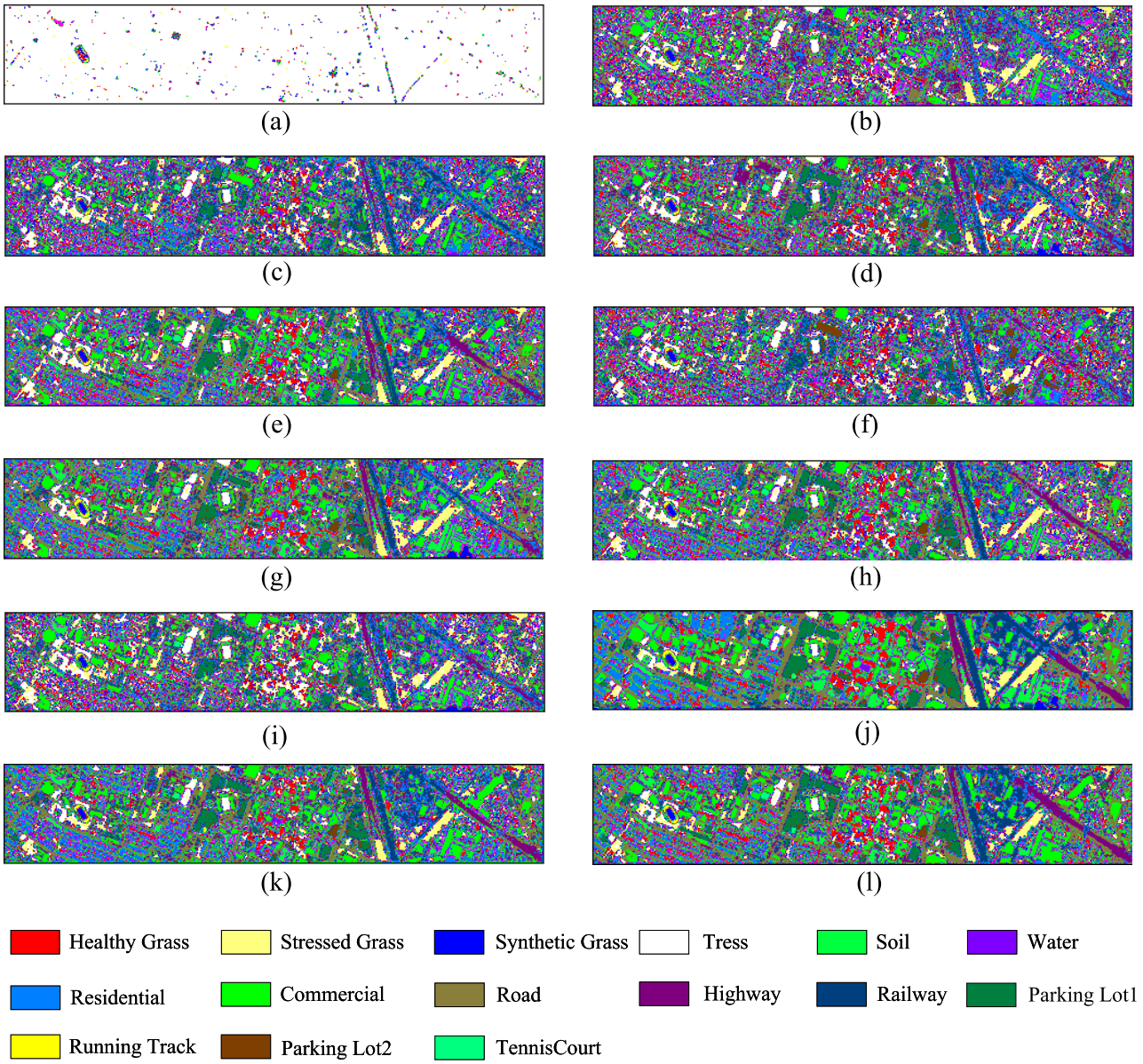


Fig. 10. Classification visualization maps of all methods on the HU2013 dataset. (a) Real ground feature map. (b)–(l) Classification map of SVM, 1D-CNN, 2D-CNN, SSRN, ViT, iFormer, GAHT, SpectralFormer, MVAHN, morphFormer, and CACFTNet (ours), respectively.

classes that are prone to misclassification in each dataset. In the IP dataset, there are classes with highly similar spectral profiles, such as the three subclasses of "corn" and three subclasses of "soybean." This similarity poses a significant difficulty in accurately distinguishing these classes during classification tasks. It can be observed that in Table VIII, most of the comparative algorithms exhibit relatively poor classification performance for the top three classes (subclasses of "corn"). On the other hand, our proposed CACFTNet method incorporates the UPCA module, which effectively explores the degree of correlation between different bands in HS images. This enables the network to focus on the bands that contribute more significantly to the classification task, resulting in better classification results for these subclasses. Consequently, the accuracy of classification for these three classes is the highest among all. Compared to the SpectralFormer method, our approach achieved significantly higher accuracies for the classes "Corn Notill," "Corn Mintill," and "corn," with improvements of 36.20% (59.68→95.88), 19.26% (80.48→99.74), and 14.68% (85.32→100.00), respectively. In addition, there are instances of imbalanced classification results. For example, the "Alfalfa" class performed poorly in the SpectralFormer method due to the relatively small number of samples. In contrast, our method achieved 100% accuracy for this class.

In Table IX, it can be observed that CACFTNet achieved the highest classification accuracy in the three classes "Meadows," "Bitumen," and "Bricks" on the PU dataset, surpassing the results obtained by other comparative algorithms. In addition, the classes "Asphalt" and "Bitumen" are quite similar, which presents a challenge in classification. Among the transformer-based comparative methods, iFormer exhibited the lowest accuracy for the class "Bitumen," while CACFTNet achieved 100% accuracy in identifying this class. For the MVAHN

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method, although its OA, AA, and kappa values are very close to our method, CACFTNet achieved optimal accuracy in the classification of three individual classes. Moreover, the values of OA, AA, and kappa for the MVAHN method differ from our method by 0.84%, 1.2%, and 1.14%, respectively.

In the HU2013 dataset, the classes "Highway" and "Railway" exhibit similar spectral responses. When distinguishing between these classes, CACFTNet outperforms the comparative methods. In Table X, MVAHN and morphFormer are 22.49% (75.57→98.06) and 18.05% (80.01→98.06) worse than CACFTNet in recognizing the category "Highway," respectively. As a result, our method demonstrates a significant advantage in accurately differentiating highly similar samples in benchmark datasets and achieving higher results across the three evaluation metrics.

2) Visual Evaluation: We visualize the classification results obtained from the experiments on the three datasets, as shown in Figs. 8–10. In addition, from the classification graphs, we can also see that the proposed network structure is relatively stable for a large number of categories or a small number of categories. In contrast, the classification graphs of the traditional algorithms and CNN-based classification algorithms carry too much noise.

On the IP dataset, all the compared methods classify the middle white and purple blocks very roughly, only our method recognizes their edges to a greater extent in a detailed way. For the PU dataset, identify the long green block in the middle, our method looks more similar to the labeled graph. Meanwhile, for the HU2013 dataset, the dark blue blob in the image is a more difficult region to recognize, and almost all the compared methods do not recognize its contour as well as its surrounding region very completely, only our method restores the contour of its blue region as well as its surrounding region to a great extent. Therefore, our proposed CACFTNet occupies a greater advantage in dealing with edge details and further validates the effectiveness of our method in terms of classification performance.

V. Conclusion

In the current work, a CACFTNet network is recommended to boost the effectiveness of HS image classification. This approach merges the global modeling abilities of the transformer and the local information perception capability of CNN at the initial stage of feature extraction to obtain shallow spatial–spectral semantic information of HS images by means of feature fusion that sums up the feature information of two different scales. These features are then fed into the UPCA module to fully capture the similarity information between different bands, allowing attention to be paid to the more important part of the spectral band information. Next, the CAAF module with a five-layer encoder, which is able to model the abstract semantic features, fuses the more useful information with the cross-layer feature information to make the information richer. Finally, the classification results are realized through the linear layer. It has been shown through extensive experiments that our method is more efficient in realizing HS image classification.

In the future, we consider extracting more discriminative spatial–spectral features from the perspective of 3D-CNN. And use different and advanced techniques to make modifications to the transformer model architecture. For example, starting from different angles of attention (channel attention and spatial attention) and fusing them to extract feature information of HS images. In addition, the connection structure in the CAAF module and how to fuse it with attention more skillfully are also issues that should be considered in our future work.

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